

Jaynes-Cummings interaction with a traveling light pulse

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The Jaynes-Cummings model provides a simple and accurate description of the interaction between a two-level quantum emitter and a single mode of quantum radiation. Due to the multimode continuum of eigenmodes in free space and in waveguides, the Jaynes-Cummings model should not be expected to properly describe the interaction between an emitter and a traveling pulse of quantum radiation. In this article, we review a cascaded quantum system approach that accurately describes the interaction of a quantum system with an incident quantum pulse of radiation. When applied to the two-level system, different iterations of this approach lead to master equations with different Jaynes-Cummings-like Hamiltonians and damping terms. © 2024 Optica Publishing Group

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1. INTRODUCTION

In their seminal article, “Comparison of quantum and semiclassical theories with application to the beam maser,” Jaynes and Cummings outlined the formalism and basic mechanisms of resonant light-matter interactions [1]. In the quantum theory, the field and its quantum states can be expanded on multimode number states, $\phi(n_1, n_2, \dots)$ with n_i photons in the i th field mode (using the notation of [1]). Absorption and emission processes by a single two-level system (TLS) are characterized by matrix elements of the dipole operator between the emitter’s ground and excited states and of the electric field operator between field states differing in photon number in any of the field modes. The quantized electric field operator can be written as a linear combination of the annihilation and creation operators $\hat{a}_i, \hat{a}_i^\dagger$ for the different field modes, and hence couples $\phi(n_1, n_2, \dots, n_i, \dots)$ to the states $\phi(n_1, n_2, \dots, n_i + 1, \dots)$ and $\phi(n_1, n_2, \dots, n_i - 1, \dots)$ with strengths $\propto \sqrt{n_i + 1}$ and $\sqrt{n_i}$, respectively.

The Jaynes-Cummings model (JCM) is the restriction of the light-matter interaction problem to the case of a TLS coupled to a single mode of the quantized field. The JCM applies to the case of a TLS residing inside, or traveling through, an optical cavity that supports a discrete cavity mode that is nearly resonant with the TLS transition (and it typically assumes that all other cavity modes are widely detuned and hence not coupled to the TLS). The JCM has been implemented with atoms and optical cavities

[2,3], Rydberg atoms and microwave cavities [4–6], superconducting qubits and various microwave resonator designs [7,8], as well as with superconducting qubits and acoustic resonators (phonons) [9–11].

Using Pauli spin operators $\hat{\sigma}^\pm$ to represent excitation and deexcitation of the TLS, the resonant Jaynes-Cummings Hamiltonian reads

$$\hat{H}_{\text{JCM}} = i\chi(\hat{a}^\dagger \hat{\sigma}^- - \hat{a} \hat{\sigma}^+). \quad (1)$$

We note that $\hat{H}_{\text{JCM}}$ acts separately within two-dimensional subspaces $\{|n, g\rangle, |n-1, e\rangle\}$ in which the dynamics periodically exchanges a single quantum between the cavity field mode and the TLS at frequency $\chi\sqrt{n}$. If different number states are populated, the quantum is exchanged at different frequencies leading to so-called collapses and revivals in the oscillating TLS excited state population [12].

If we want to describe the case of an atom flying through a cavity and thus traversing the standing wave field mode as depicted in Fig. 1(a), the JCM Hamiltonian Eq. (1) merely has to be equipped with a time-dependent value $\chi(t)$ reflecting the amplitude of the field mode at the time-dependent position of the atom.

A natural question arises: can we employ a similar time-dependent Jaynes-Cummings Hamiltonian,

$$\hat{H}_{\text{JC-pulse}}(t) \stackrel{?}{=} i\sqrt{\gamma}u(t)(\hat{a}^\dagger \hat{\sigma}^- - \hat{a} \hat{\sigma}^+), \quad (2)$$

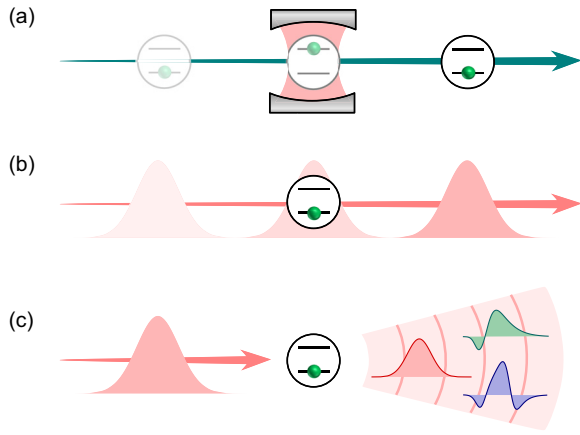


Fig. 1. (a) A two-level atom passing through a single-mode cavity. (b) A single-mode pulse traveling over a two-level atom. (c) A light pulse interacting with a two-level system leads to emission of photons into a continuum of field modes.

if a stationary TLS is coupled to an incident pulse of radiation with amplitude $\sqrt{\gamma}u(t)$, cf., Fig. 1(b)? Certainly, photons may be absorbed and emitted by the TLS like in the case of a cavity, but the propagation of the wave packet through space explores a continuum of wave number and frequency components, which are close to resonance and hence also available for the photons emitted by the TLS. A single photon may thus be absorbed from the pulse and be subsequently re-emitted into wave packets of different temporal shapes, as depicted in Fig. 1(c), and, due to the saturation non-linearity of the TLS, scattering of a multiphoton input state may feed many output modes with different number state contents. Due to its assumption of a strictly single-mode field, the successes of the JCM within the fields of cavity QED and circuit QED seem not to extend to scattering of traveling pulses on a TLS, i.e., to the field of waveguide QED.

There has been recent progress on this problem coming from different perspectives, and we shall here review a particular cascaded system approach that represents the pulse of quantum radiation as the emission from a leaking cavity field mode. In this article, we review this novel approach with particular emphasis on the emerging Jaynes-Cummings interaction between the leaking cavity mode and the TLS. Several different forms of the JCM with damping terms can, indeed, be used to describe the interaction of traveling pulses of radiation with a TLS. The article is organized as follows. In Section 2 we review general formalism for the description of quantized light-matter interactions. Explicit analytical solutions exist for the scattering of wave packets with a single or two excitations on a TLS, while the large dimension of the multiphoton multimode Hilbert space becomes computationally challenging for only a few more photons.

In Section 3, we review our recent cavity mode representation of the traveling wave packet pulse and the resulting cascaded cavity-TLS master equation. In this equation, the Hamiltonian is of the same form as Eq. (2), but with a time dependence that differs from the pulse shape that we try to model. The field leaking from the cavity and from the TLS interfere in the forward scattered radiation continuum and is represented by a correlated damping term in the master equation.

In Section 4, we turn to the quantum state of the scattered light. In the cascaded systems approach, this is made possible by restricting the analysis of the output field continuum to a single or a small number of output temporal modes. Our theoretical description at this stage comprises the input mode, the TLS, and one or several output mode oscillators, which are also coupled to the TLS by effective time-dependent JCM Hamiltonians. This analysis permits evaluation of the final quantum state of any output mode and hence of protocols for transforming quantum states of light by the non-linear interaction with a TLS. We review genuine multimode capabilities that are by their nature not present in the single-mode JCM but are revealed by the cascaded systems calculations. Useful capabilities include high fidelity sorting of different number states into different modes [13] and subtraction and transfer of a single photon from an input pulse to an orthogonal pulse mode [14,15].

In Section 5, we adopt an interaction picture, where the TLS couples to time-dependent linear combinations of the input and output modes. This interaction picture leads to a numerically efficient treatment of fields with many photons and in the special case of scattering on a TLS, we recover the intuitive time-dependent Jaynes-Cummings-like Hamiltonian proposed in Eq. (2). The Jaynes-Cummings interaction is in the interaction picture now supplemented by the coupling of the TLS to another orthogonal mode combination, and inclusion in the master equation of their joint loss to the continuum of modes.

In Section 6, we present three examples of how the most characteristic behaviors of the Jaynes-Cummings model are recovered or modified in the case of the interaction with a traveling pulse: Rabi oscillations, collapses and revivals, and photon subtraction.

Section 7 concludes with a summary of the main results of the article.

2. QUANTUM LIGHT-MATTER INTERACTION

A natural starting point to describe the quantum interaction between a propagating pulse and a TLS, but *not* the one adopted in this work, is by introduction of a general representation of the quantum state of the radiation and its accompanying Schrödinger equation. In a one-dimensional (1D) waveguide, the continuous field operators $\hat{a}(t)$ evaluated at a fixed location satisfy the commutation relation $[\hat{a}(t), \hat{a}^\dagger(t')] = \delta(t - t')$, and a pure state of the field can be expanded in the form

$$|\psi\rangle = \sum_n \int dt_1 \cdots dt_n \psi_n(t_1, \dots, t_n) \hat{a}^\dagger(t_1) \cdots \hat{a}^\dagger(t_n) |0\rangle, \quad (3)$$

where $\psi_n(t_1, \dots, t_n)$ denotes the n -photon wave function. While the scattering of one- and two-photon states on a single emitter can be solved by scattering theory [16–20], conveniently formulated as an input-output network theory [21–23], the complex (many-body) character of the quantum state of the field makes the interaction of just a few more photons with a TLS prohibitively complicated to solve. Despite that several general formalisms have been established for an arbitrary number of photons [24–29], their numerical implementation is restricted to the few-photon regime due to the complexity of the general problem, as illustrated by the numerical procedures

presented in [28]. A remarkable exception is the calculation by Fischer [30] of the final state amplitude on the original state and on the state with a single quantum transferred from the TLS to an exponentially decaying wave packet mode.

Even in the case where the initial state of the incident field occupies only a single temporal mode $u(t)$ with creation operator $\hat{a}_u^\dagger = \int dt u(t) \hat{a}^\dagger(t)$, the exchange of quanta of excitation with the TLS may readily lead to population of a correlated state of the general form Eq. (3).

If we should be able to obtain the quantum state of the emitted field in the form of Eq. (3), we may consider a simpler characterization of the scattered field, i.e., the quantum state of photons in a single temporal mode $v(t)$, described by the annihilation operator $\hat{a}_v = \int dt v^*(t) \hat{a}(t)$. The density matrix for such a single-mode mixed state can be obtained by tracing out the modes orthogonal to $\hat{a}_v$, $\rho_v = \text{Tr}_{v^\perp} [|\psi\rangle\langle\psi|]$, and is given by the formal expression

$$\rho_v = \sum_{m=0}^{\infty} \frac{1}{m!} \int dt_1 dt_2 \cdots dt_m \langle 0 | \hat{b}(t_m) \cdots \times \hat{b}(t_2) \hat{b}(t_1) | \psi \rangle \langle \psi | \hat{b}^\dagger(t_1) \hat{b}^\dagger(t_2) \cdots \hat{b}^\dagger(t_m) | 0 \rangle, \quad (4)$$

where $\hat{b}(t) = \hat{a}(t) - v(t) \hat{a}_v$. The reduced density matrix ρ_v is the quantity of interest for many quantum optical applications, and it permits, for example, direct experimental distillation via the quantum pulse gate [31,32].

Using input-output theory, it has been shown that for a single-mode input field the dynamics of the TLS, can be obtained from a hierarchy of coupled master equations [33]. This theory does not provide the quantum state of the output field, but it makes it possible to obtain mean values and correlation functions and to simulate the outcome of continuous measurements on the output field [34–36].

All the methods applicable to interactions between quantum systems and pulses thus seem to be very different even from the spirit of the Jaynes-Cummings model, but in Section 3, we review a method to cast the problem in such a way from the beginning that we can employ a cascaded-system master equation [37,38]. For the interaction with a TLS, this leads to a master equation containing the celebrated Jaynes-Cummings Hamiltonian and, in this article we shall summarize some of the distinctive differences and similarities between the interaction with traveling pulses and standing waves in a cavity. In Section 4, we show that the reduced density matrix ρ_v of the field in any single-output mode v can be obtained directly and easily from an extended version of the JCM and cascaded master equation, i.e., without ever computing the full output state $|\psi\rangle$ in Eq. (4).

3. CASCADED SYSTEM DESCRIPTION OF A QUANTUM PULSE INCIDENT ON A TWO-LEVEL SYSTEM

We consider a pulse of radiation with a spatial amplitude $u(x, t)$ incident on the TLS. The pulse is assumed to move at a constant speed so it is equivalently specified by its spatial amplitude at a fixed early moment in time and by the time-dependent amplitude $u(t)$ at a fixed position in space.

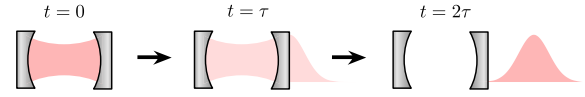


Fig. 2. Illustration of our physical modeling of a traveling pulse as the emission from a single-mode leaking cavity. For a linear loss process, the output field is a single traveling mode copy of the initial cavity mode quantum state.

In [39–41], we argued that the interaction with the pulse is equivalent to the one obtained by positioning the TLS at a distance downstream from a cavity that emits its quantum state into a pulse of the shape $u(t)$, as depicted in Fig. 2. For now, we assume unidirectional propagation of the field, and the constant propagation delay due to their separation Δx can be incorporated by a formal change of time argument as if the state assigned to the cavity mode at time t in reality applies at the earlier time $t - \Delta x/c$. The resulting equation treats the TLS as if it is located in the immediate vicinity outside the cavity, as shown in Fig. 3(a).

To leak a pulse with temporal shape $u(t)$ and any desired Fock state or pure or mixed quantum state, the cavity mode in Fig. 2 must be initiated in the desired quantum state and it must be subject to the coherent out-coupling amplitude [39]

$$g_u(t) = \frac{u^*(t)}{\sqrt{1 - \int_0^t dt' |u(t')|^2}} \quad (5)$$

(see also [16,42]).

According to the theory of cascaded quantum systems [37,38], this amplitude governs the Jaynes-Cummings-like interaction Hamiltonian between the cavity emitter mode and the TLS,

$$\hat{H}_{\text{JCM-I}}(t) = \frac{i}{2} (\sqrt{\gamma} g_u(t) \hat{a}^\dagger \hat{\sigma}^- - \sqrt{\gamma} g_u^*(t) \hat{a} \hat{\sigma}^+), \quad (6)$$

while the ultimate leakage of all excitation from the two systems into the waveguide is described by master equation damping terms.

Downstream from the TLS, the radiation field is composed of the vacuum input to the waveguide, $\hat{b}_{\text{in,vac}}(t)$, the field amplitude, $g_u(t) \hat{a}$, leaked by the cavity and the amplitude, $\sqrt{\gamma} \hat{\sigma}^-$, leaked by the TLS. These terms interfere, and downstream detection of a photon in the waveguide is accompanied by the action of the collective annihilation operator,

$$\hat{b}_{\text{out}}(t) = \hat{b}_{\text{in,vac}}(t) + g_u^*(t) \hat{a}_u + \sqrt{\gamma} \hat{\sigma}^-, \quad (7)$$

on the joint state of the input vacuum modes, the cavity mode, and the TLS.

The operation of $\hat{b}_{\text{in,vac}}(t)$ on vacuum is trivial, and the radiation loss represented by $\hat{b}_{\text{out}}(t)$ corresponds to the action of the jump operator $\hat{L}_{\text{JCM-I}} = g_u^*(t) \hat{a}_u + \sqrt{\gamma} \hat{\sigma}^-$ in the Lindblad master equation for the joint TLS and single-cavity-mode density matrix:

$$\frac{d\rho}{dt} = -i[\hat{H}_{\text{JCM-I}}, \rho] + D[\hat{L}_{\text{JCM-I}}]\rho, \quad (8)$$

where $D[\hat{L}]\rho \equiv -\frac{1}{2}(\hat{L}^\dagger \hat{L} \rho + \rho \hat{L}^\dagger \hat{L}) + \hat{L} \rho \hat{L}^\dagger$.

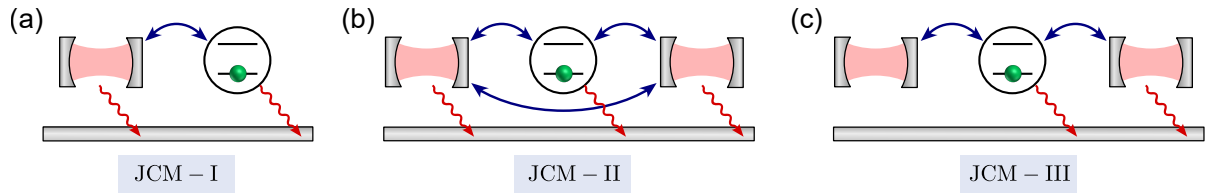


Fig. 3. (a) Illustration of our effective model for describing the interaction between a two-level system and an input pulse; see Eq. (8). (b) Illustration of our model permitting analysis of the output state in any specified temporal mode; see Eq. (13). (c) When the output temporal mode [cf. panel (b)] is chosen identical to the input pulse mode, we can recover a particularly efficient description in the interaction picture; see Eq. (24). The bent and curvy arrows represent the Jaynes-Cummings-like interactions given by Hamiltonians $\hat{H}_{\text{JCM-I(II)(III)}}$ and the interfering emission into the waveguide, given by the Lindblad damping operators $\hat{L}_{\text{JCM-I(II)(III)}}$. The combination of the Hermitian interaction and the damping terms causes excitation to only propagate from left to right in panels (a) and (b).

Equation (8) resembles a damped version of the Jaynes-Cummings model, but our system differs from the conventional Jaynes-Cummings dynamics in two important ways: (i) the coupling term has a very different time dependence than suggested in Eq. (2), and (ii) after the passage of the pulse, both the (upstream) oscillator mode and the TLS have lost all excitations into downstream propagation of radiation away from the cavity and TLS.

As a further interesting deviation from the usual JCM dynamics, we note that terms in the master equation coming from the unitary commutator part and from the Lindblad anticommutator terms $\frac{1}{2}(\hat{L}^\dagger \hat{L} \rho + \rho \hat{L}^\dagger \hat{L})$ conspire so that only $\hat{a} \hat{\sigma}^+$ ($\hat{a}^\dagger \hat{\sigma}^-$) acts on ρ from the left (right). This implies the cascaded nature of the problem: no excitation propagates upstream from the TLS to the oscillator mode.

4. QUANTUM STATE OF THE SCATTERED LIGHT

Since the downstream light field is treated as loss, our master equation does not reveal its final quantum state in the same way as time evolution by the Hamiltonian Eq. (1) explicitly provides the state of a cavity field mode at any time during and after the Jaynes-Cummings interaction with a TLS.

The downstream radiation may be multimode and hence it may be prohibitively complicated to obtain its exact quantum state in the form of Eq. (3). There is, however, no obstacle to the calculation of the quantum state contents of any specified single wave packet mode $v(t)$ in the output field. To that end, we can apply a similar cavity trick as above and assume a downstream cavity with a coupling amplitude $g_v(t)$ to the waveguided field, which would fully capture the field populating the wave packet mode $v(t)$, i.e., the reverse of the process that we employed for emission of $u(t)$. Such a cavity should couple to the waveguide by the coupling strength

$$g_v(t) = -\frac{v^*(t)}{\sqrt{\int_0^t dt' |v(t')|^2}}. \quad (9)$$

The cascaded system of the emitter cavity, the TLS, and the final pick-up cavity is depicted in Fig. 3(b). Its unidirectional dynamics is described by a Hamiltonian that exchanges quanta between all three components with their respective coupling strengths to the waveguide:

$$\begin{aligned} \hat{H}_{\text{JCM-II}}(t) = & \frac{i}{2} (\sqrt{\gamma} g_u(t) \hat{a}_u^\dagger \hat{\sigma}^- + \sqrt{\gamma} g_v^*(t) \hat{\sigma}^+ \hat{a}_v \\ & + g_u(t) g_v^*(t) \hat{a}_u^\dagger \hat{a}_v - \text{H.c.}). \end{aligned} \quad (10)$$

The field components that are not retrieved by the pick-up cavity are governed by the annihilation operator

$$\hat{b}_{\text{out}}(t) = \hat{b}_{\text{in,vac}}(t) + g_u^*(t) \hat{a}_u + \sqrt{\gamma} \hat{\sigma}^- + g_v^*(t) \hat{a}_v. \quad (11)$$

This residual loss of radiation is represented by the Lindblad damping term,

$$\hat{L}_{\text{JCM-II}}(t) = \sqrt{\gamma} \hat{\sigma}^- + g_u^*(t) \hat{a}_u + g_v^*(t) \hat{a}_v, \quad (12)$$

in the master equation for the joint state of our quantum system that now consists of the TLS and two cavity modes representing the input pulse and a selected output pulse:

$$\frac{d\rho}{dt} = -i[\hat{H}_{\text{JCM-II}}(t), \rho] + D[\hat{L}_{\text{JCM-II}}(t)]\rho. \quad (13)$$

The scattering of a quantum pulse is calculated by assuming any initial quantum state of the input cavity mode, u , and the ground and vacuum states of the TLS and pick-up cavity mode, v . After solving master Eq. (11), the TLS and input cavity are in the ground state and the reduced density matrix ρ_v of the pick-up cavity mode represents the quantum state of the outgoing wave packet $v(t)$, as in Eq. (4).

In Eqs. (5) and (9) the coupling amplitudes formally diverge in the $t \rightarrow \infty$ and $t \rightarrow 0$ limits, respectively. At these times the cavities are empty, and the relevant product of state amplitudes and coupling strengths should be finite. To avoid numerical errors, we suggest to add a small regularizing term $\delta n \ll 1$ (e.g., $\delta n \sim 10^{-4}$) within the square root terms in both expressions. This corresponds to permitting a residual photon occupation of such magnitude, and it does not alter the physical solutions in any appreciable way.

5. INTERACTION PICTURE WITH RESPECT TO THE FREE PROPAGATION OF A SINGLE-MODE PULSE

We have established a Hamiltonian Eq. (10) and a master Eq. (13) that describe the interaction of an incoming pulse of quantum radiation with a two-level quantum system, as well as the state of the field retrieved in any single mode of the output field. This description deals with two field modes rather than

the single mode in the Jaynes-Cummings model. In this section, we will show that it is possible to recover a description where the hypothesized Jaynes-Cummings Hamiltonian Eq. (2) is the dominant interaction term, and where extra components take care of the multimode nature of the problem.

To achieve this simple description, we perform a transformation to the interaction picture defined by the cavity-cavity coupling part of the Hamiltonian

$$\hat{H}_0(t) = \frac{i}{2} [g_u(t)g_v^*(t)\hat{a}_u^\dagger\hat{a}_v - g_v(t)g_u^*(t)\hat{a}_v^\dagger\hat{a}_u]. \quad (14)$$

At this point, we assume that the input and output virtual cavities address the same temporal shape $u(t) = v(t)$, while the general case is described in [41]. The equations of motion for the operators in the interaction picture are

$$\dot{\hat{a}}_{u,\text{IP}}(t) = \frac{1}{2}g_u(t)g_v^*(t)\hat{a}_{v,\text{IP}}(t), \quad (15)$$

$$\dot{\hat{a}}_{v,\text{IP}}(t) = -\frac{1}{2}g_u^*(t)g_v(t)\hat{a}_{u,\text{IP}}(t), \quad (16)$$

where IP denotes that the operator is described in the interaction picture. The solution to this coupled set of equations is

$$\hat{a}_{u,\text{IP}}(t) = \cos \theta(t)\hat{a}_u(0) - \sin \theta(t)\hat{a}_v(0), \quad (17)$$

$$\hat{a}_{v,\text{IP}}(t) = \cos \theta(t)\hat{a}_v(0) + \sin \theta(t)\hat{a}_u(0), \quad (18)$$

where $\theta(t)$ is determined by the condition

$$\frac{d}{dt}\theta(t) = -\frac{1}{2}g_u(t)g_v^*(t). \quad (19)$$

Our assumption that $v(t) = u(t)$ guarantees that $\theta(t)$ is real, and we can show that

$$\sin^2 \theta(t) = \int_0^t dt' |u(t')|^2, \quad (20)$$

by differentiating both sides of the equation with respect to time. This yields

$$2 \sin \theta(t) \cos \theta(t) \frac{d\theta(t)}{dt} = |u(t)|^2, \quad (21)$$

and we note that due to Eq. (20), the factors $\cos \theta(t)$ and $\sin \theta(t)$ are nothing but the denominators in Eqs. (5) and (9).

It is evident from the definition of $\theta(t)$ that $\sin^2 \theta(t=0) = 0$ and $\sin^2 \theta(t \rightarrow \infty) = 1$, such that during the interaction with the pulse, the cavity operators are completely exchanged. In the interaction picture the u -cavity operator “follows the pulse”, and in the case of no intermediate emitter, it represents the constant quantum state content of the pulse at all times. When the scatterer is present, the transformed Hamiltonian in the interaction picture becomes

$$\begin{aligned} \hat{H}_{\text{JCM-III}}(t) &= i\sqrt{\gamma}u(t)(\hat{a}_{u,\text{IP}}(t)^\dagger\hat{\sigma}^- - \hat{\sigma}^+\hat{a}_{u,\text{IP}}(t)) \\ &+ i\sqrt{\gamma}u(t)\cot 2\theta(\hat{a}_{v,\text{IP}}(t)^\dagger\hat{\sigma}^- - \hat{\sigma}^+\hat{a}_{v,\text{IP}}(t)), \end{aligned} \quad (22)$$

which contains in the first line precisely the naive guess, hypothesized in Eq. (2) in Section 1. Our analysis, however, also yields the interaction with a second mode, which is initially the empty downstream cavity mode. This mode gets transiently occupied and hence forms a non-Markovian element in the TLS dynamics in the interaction picture, represented by the rightmost cavity in Fig. 3(c). The pulse mode suffers no direct loss, while excitation transferred to the TLS and subsequently to the second mode leaks as revealed by evaluating the Lindblad operator in the interaction picture

$$\hat{L}_{\text{JCM-III}}(t) = \sqrt{\gamma}\hat{\sigma}^- - (\tan \theta + \cot \theta)u(t)\hat{a}_{v,\text{IP}}(t). \quad (23)$$

To summarize, we have found that the interaction between a traveling pulse and a quantum emitter has an intuitive Jaynes-Cummings-like component, which is supplemented with loss and coupling to a secondary, lossy mode in the interaction picture master equation

$$\frac{d\rho}{dt} = -i[\hat{H}_{\text{JCM-III}}(t), \rho] + D[\hat{L}_{\text{JCM-III}}(t)]\rho. \quad (24)$$

6. RESULTS

In the previous sections, we have reviewed an efficient description of the interaction between a TLS and a traveling pulse mode of quantum radiation. We have shown that despite the multimode character of propagating fields, it is possible to recast the problem with a Jaynes-Cummings-like interaction Hamiltonian and Lindblad loss terms acting on only the TLS and one or two oscillator modes.

Our analysis has identified three different master equations, Eqs. (8), (13), and (24). In this section, we present solutions of these equations, and we show how the interaction between a TLS and a traveling pulse has similarities with and deviates from the time-dependent interaction with a resonant cavity mode.

Numerical results in this section are obtained by use of the QuTiP toolbox [43,44].

A. Rabi Oscillations with a Number State Pulse

In master Eq. (8), the coupling strength $g_u(t)$ differs from the shape $u(t)$ of the pulse. Note, however, that $\hat{H}_{\text{JCM-I}}$ acts on a quantum state with progressively fewer photons, governed by the same factor as the denominator in Eq. (5), and hence its strength follows the value of $u(t)$ in an average sense.

Figure 4(a) shows the photon number and TLS excitation probability according to the unitary JCM dynamics of Eq. (1) with a time-dependent coupling, $\chi(t) = \sqrt{\gamma}u(t)$, with a Gaussian temporal width $\tau = \gamma^{-1}$, for the cavity mode initially in an $|n=20\rangle$ Fock state. The figure also shows the solution of the master equation incorporating decay from the excited state of the TLS with rate γ , where we note an accumulated loss of a little more than two photons.

Figure 4(b) shows the results for an incident Gaussian wave packet, $u(t)$, with an $|n=20\rangle$ Fock state of radiation scattering on the TLS, as in Eq. (8). The TLS excited state population yields similar damped Rabi oscillations as in Fig. 4(a), accompanying the rapid release of the field from the u -cavity. Figure 4(b) also shows the population emerging in the outgoing wave packet

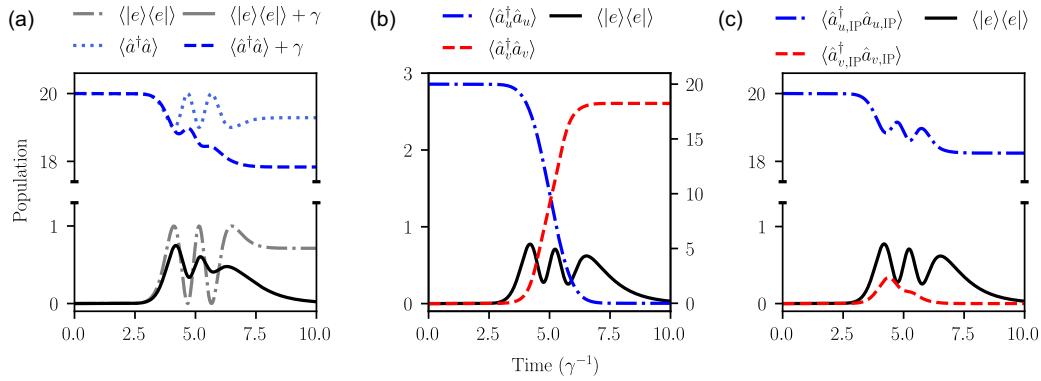


Fig. 4. Rabi oscillations of a TLS interacting with an $|n = 20\rangle$ Fock state. Panel (a) shows the TLS excitation dynamics (lower curves) and the mean number of oscillator quanta (upper curves) for the unitary JCM with Gaussian time-dependent coupling $\chi(t) = \sqrt{\gamma}u(t)$, with $\tau = \gamma^{-1}$, and for the model with a TLS with decay rate γ . Panel (b) shows the results of Eqs. (8) and (13) for a traveling pulse with the same Gaussian form of $u(t)$ as in panel (a). The TLS excitation dynamics is similar but differ from that in panel (a), while the excitation leaks from the upstream u -cavity, and is partly recovered in the downstream v -cavity. Note the left y -axis pertains to the TLS and the right y -axis pertains to the cavity excitation dynamics. Panel (c) shows the same results but for the interaction picture modes applied in Eq. (24). The u_{IP} -mode represents the traveling pulse and its excitation dynamics is similar to the one of the oscillator mode in the damped JCM, shown in panel (a). The ancillary v_{IP} -mode is weakly excited and causes the small quantitative discrepancies with the results in panel (a).

$v(t)$ (chosen here on the same form as $u(t)$) after the interaction with the TLS, as in Eq. (13). We note a loss of a little less than two quanta of radiation to other modes.

Figure 4(c) shows the same TLS excitation dynamics but calculated according to Eq. (24). The interaction picture annihilation operator $\hat{a}_{u,\text{IP}}(t)$ represents the $u(t)$ wave packet mode both before and after the interaction with the TLS, and its population dynamics (upper dotted-dashed curve) resembles the conventional JCM dynamics in presence of TLS decay, shown in Fig. 4(a). The interaction picture annihilation operator $\hat{a}_{v,\text{IP}}(t)$ represents an orthogonal mode that is weakly excited and thus modifies the TLS dynamics during the interaction.

Although we obtain results showing similar features with the single-mode time-dependent JCM with damping and with our modified JCM-like master equations, there are quantitative discrepancies. These are due to the multimode character of the non-linear scattering of traveling pulses, which is only taken fully into account in the extended versions of the JCM presented here.

B. Number State versus Coherent State Dynamics: Collapses and Revivals

The cascaded nature of master Eqs. (8) and (13) ensures that the combination of the commutator and the Lindblad anti-commutator terms is responsible for excitations propagating only downstream (as only annihilation and no creation operators for the upstream cavity act on ρ from the left, and vice versa from the right). This has another striking consequence: if the cavity field mode is initially in a coherent state with amplitude α_0 , it remains a (damped) coherent state $|\alpha(t)\rangle$ for all later times. The TLS thus evolves as if driven by a classical pulse with a time-dependent field amplitude $\alpha(t)g_u^*(t) = \alpha_0 u(t)$ [34,40]. In addition, the TLS decays to the waveguide with the rate γ , and the system may thus show damped Rabi oscillations but no trace of the collapse and revival phenomenon, which is a crucial feature of the interaction with a coherent state in a cavity mode.

Conversely, for an initial number state, which would exhibit perfect Rabi oscillations in the JCM, our Eq. (8) evolves the system through a distribution of excitation subspaces with a corresponding distribution of Rabi oscillation frequencies $g\sqrt{n}$. This could suggest the occurrence of collapses and revivals in the TLS excited state population, which would be the complete reverse of the roles of coherent states and number states in the JCM.

While we do observe Rabi oscillations, despite the structure of Eq. (8), we see no signs, however, of collapses and revivals in the numerical solutions in Fig. 4(b). We ascribe this to the role played by the dissipation terms in the master equation and note that the absence of the collapses and revivals may also be inferred from the equivalence between Eqs. (8) and (24), and the fact that the field mode in the latter equation does not display the similarly large uncertainty in the number of quanta.

C. Further Dissipation, Non-chiral Coupling

In the previous paragraphs, we have mentioned the possibility of incorporating decay and dissipation in the JCM through damping terms in a master equation. Similarly, it is straightforward to supplement our master equations with further damping terms of the same form $D[\hat{L}_i]\rho$ as the Lindblad terms already present in Eqs. (8), (13), and (24).

This is not the place for an exhaustive review of such damping terms, but let us consider here what may have seemed a severe restriction of our analyses—a perfect unidirectional interaction between the TLS and the propagating fields. Most emitters couple to fields propagating in both directions along waveguides, and if we want to incorporate the possibility of reflection from the emitter, it is only a matter of adding a corresponding Lindblad term $D[\hat{L}_{\text{refl}}]\rho$ with $\hat{L}_{\text{refl}} = \sqrt{\gamma'}\hat{\sigma}^-$ to our master equations. The analysis of the TLS dynamics and the analysis of the forward propagating field proceed by solving the master equations with this extra term.

In Fig. 5(a), we show a particular process that occurs in the case of perfect chiral coupling of a TLS to a two-photon wave

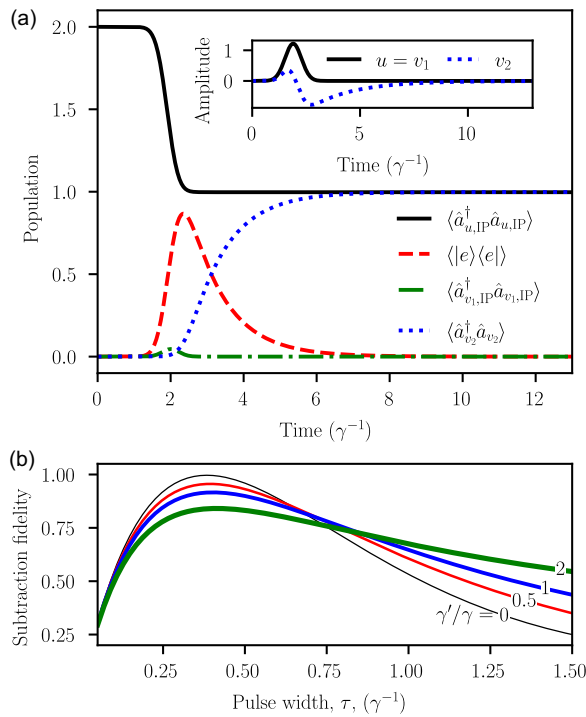


Fig. 5. Panel (a) presents the solution of Eq. (24) for the interaction of a TLS with a Gaussian pulse of width $\tau = 0.3799\gamma^{-1}$ and two quanta of excitation. The calculations show the removal of one quantum from the input mode and the emergence of one quantum in a new, orthogonal temporal mode (these modes are shown in the inset). The density matrix elements for the modes (not shown) reveal that the two output modes are, indeed, populating $|n = 1\rangle$ Fock states. Panel (b) shows the fidelity of the subtraction of a single quantum from the incident pulse as a function of the pulse duration, and for different values of the TLS emission rate in the direction opposite to the pulse propagation. Perfect subtraction only occurs in the case of perfect chiral coupling of the TLS to the traveling pulse.

packet, namely, the very precise removal of exactly one photon from the incident pulse $u(t)$ [14,15]. The dynamics is obtained in the interaction picture, using Eq. (24). Naively, this process is well understood from the JCM Rabi oscillation dynamics, causing excitation of the TLS with a π -pulse and subsequent emission of a single quantum of excitation. However, in our case, the TLS is constantly coupled to and emitting into the waveguide modes. As shown by the red curve in Fig. 5(a), the TLS is not fully excited at any time, but for the right parameter values, it emits, over time, precisely one quantum of excitation (blue dotted curve) into a temporal pulse mode $v_2(t)$, which is orthogonal to $u(t)$ (see the modes in the inset). This effective splitting of a number state pulse into two one-photon states may have useful implications in quantum information processing with light, but will it only work if we can ensure the unidirectional coupling of the TLS to the traveling fields (the assumption in [14])? The answer is given in Fig. 5(b), where we show the fidelity of the photon subtraction process as a function of the duration of the incident Gaussian two-photon pulses. The different curves are obtained for different values of the emission rate in the direction of reflection from the TLS. Only for the perfect chiral coupling do we find a pulse duration with perfect subtraction. The formalism presented here applies

also to fields populating multiple (a few) input and output modes by introduction of cascaded cavities, located upstream and downstream from the TLS [40].

7. CONCLUSION

The textbook Jaynes-Cummings model assumes a single quantized field mode and, hence, it does not describe the interaction between a two-level system and a traveling pulse of light due to the multimode character of the field continuum available for the scattered photons. We have reviewed a cascaded open-system approach that leads to master Eqs. (8), (13), and (24) of similar form and complexity as the ones describing the Jaynes-Cummings model. We remind the reader that the cavities depicted in Figs. 2 and 3 are only introduced to facilitate the theoretical description of what are propagating temporal modes in the actual experiment. It is through this effective description that we recover single- and few-mode variants of the Jaynes-Cummings interaction. With the assumption of the validity of the Born-Markov approximation of the emission of the TLS into the waveguide, our master equations are exact.

By our discussions of the different forms of the master equations and by our presentation of results for specific processes we have demonstrated the most important consequences of the modifications to the Jaynes-Cummings model in the case of propagating light. We are delighted that by so simple modifications of the theory, the 60-year-old Jaynes-Cummings model may still be counted on and provide intuition and quantitative predictions for a wide range of future applications.

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